

Organic Chemistry

• Definition:-

Study of carbon compounds. (organic compounds)

- Carbon has ability to link with itself & other atoms to form a large no. of compounds.

Catenation

self-linking ability of C

• Organic Compounds :-

- Compound that contain Carbon
 - also defined as hydrocarbons and their derivatives.
- compounds contain hydrogen & carbon only

• Hydrocarbons :-

Hydrocarbons are divided into :-

Saturated

- Hydrocarbons whose C-C bonds are all single bonds.
- Single bond = sigma bond
- also called Alkanes
- General formula = C_nH_{2n+2}
- Further molecule cannot be added - only substitution (replacement) happens.



e.g.:- Methane, Ethane, Propane.

Unsaturated

- Hydrocarbons that contain C-C multiple bonds (double or triple)
- Sigma + pi bond
- Divided into 2 types.

Alkenes

- Contain atleast 1 =
- General formula C_nH_{2n}
- e.g.:- Ethene
- Meth doesn't exist in alkenes.

Alkynes

- Contain atleast 1 $\equiv$
- General formula C_nH_{2n-2}
- e.g.:- Ethyne
- Meth doesn't exist in alkynes.

• Bond :-

	Sigma Bond	Pi Bond
Overlap	head to head	side by side
Strength	strong	Weak
Breaking	Hard to break	Easy to break
Present in	Single, double n, triple bond	Only in double n triple bond.

Single bond = 1 sigma bond

Double bond = 1 sigma + 1 pi bond

Triple bond = 1 sigma + 2 pi bond

• Substitution rxn :-

- Rxn of saturated hydrocarbons
- One atom or group replaces another atom in molecule.
- further molecule cannot be added, only replacement occurs
- Eg 1- $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$

• Addition rxn :-

- Rxn of unsaturated Hcarbons
- A molecule is added across the double or triple bond.
- Pi bond breaks (bcz Pits weak n easy to break)
- Example 1- $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_3$

• Elimination Rxn :-

Opposite of addition : a molecule is removed.

• Condensation Rxn :-

two molecules join together with loss of small molecule (water).

* Why Meth doesn't exist in unsaturated hydrocarbons?

Meth doesn't exist in unsaturated bcz

1 C cannot form a double or triple bond with itself.

* Homologous Series

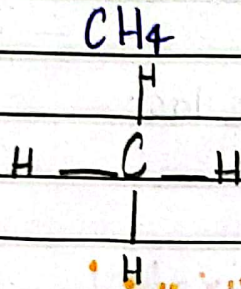
Alkanes	C_nH_{2n+2}
Alkenes	C_nH_{2n}
Alkynes	C_nH_{2n-2}
Alcohols	$C_nH_{2n+1}OH$
Carboxylic acid	$C_nH_{2n+1}COOH$

* C-Number table :-

1 - Meth	8 - Oct
2 - Eth	9 - Non
3 - Prop	10 - Dec
4 - But	
5 - Pen	
6 - Hex	
7 - Hep	

Alkanes (Saturated) General formula C_nH_{2n+2}

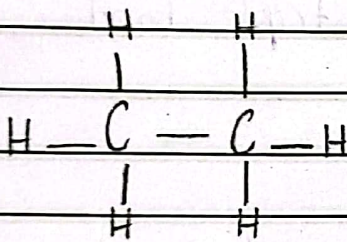
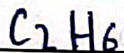
Methane



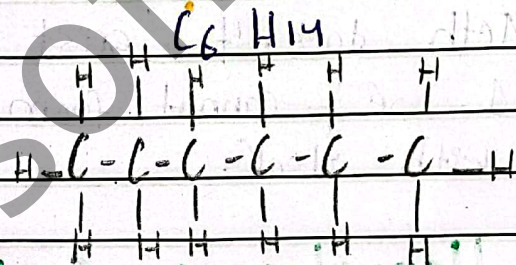
Pentane $CH_3(CH_2)_3CH_3$



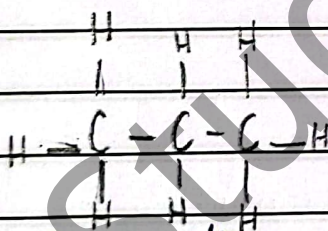
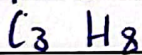
Ethane (CH_3-CH_3)



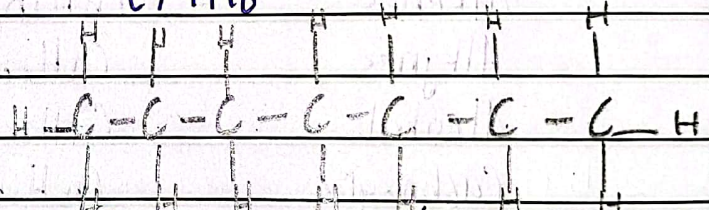
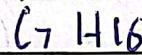
Hexane $CH_3(CH_2)_4CH_3$



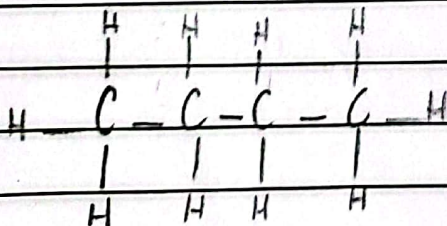
Propane $(CH_3-CH_2-CH_3)$



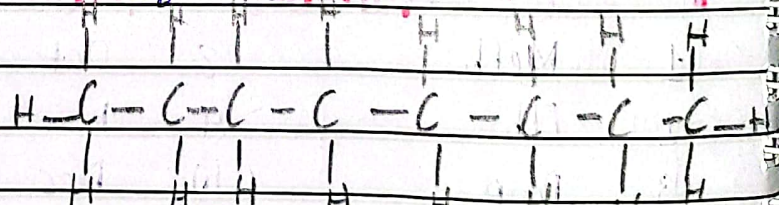
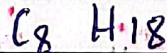
Heptane $CH_3(CH_2)_5CH_3$



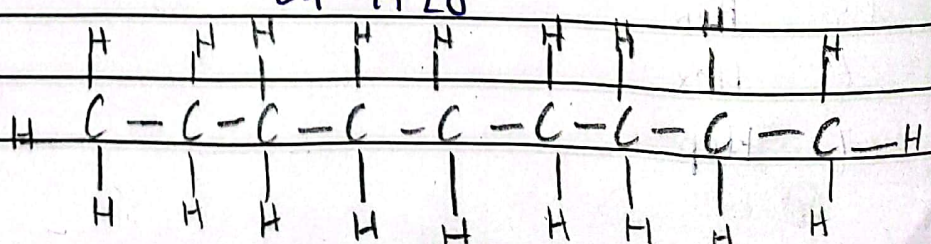
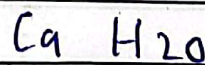
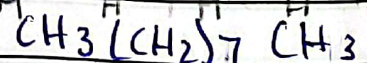
Butane $(CH_3-CH_2-CH_2-CH_3)$



Octane $CH_3(CH_2)_6CH_3$

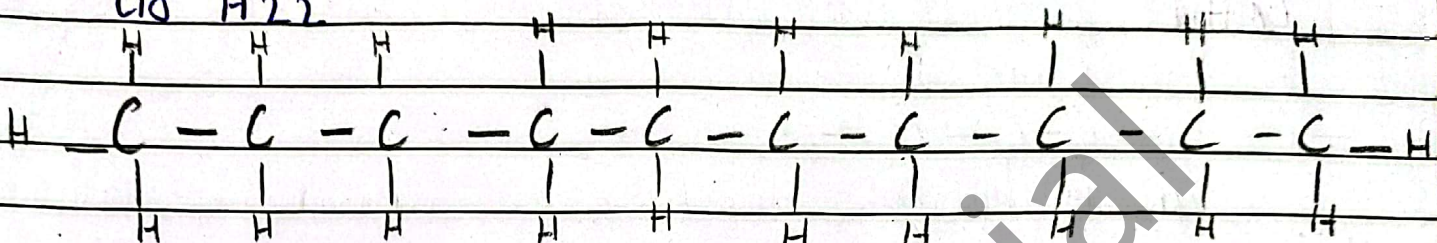


Nonane



10- Decane $\text{CH}_3(\text{CH}_2)_8\text{CH}_3$

$\text{C}_{10}\text{H}_{22}$



Alkenes

(unsaturated) (General formula C_nH_{2n})

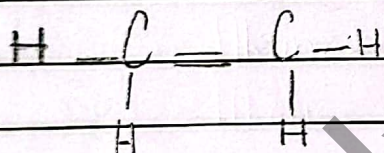
• Ethene

C_nH_{2n}

$n = 2$

$\text{C}_2\text{H}_2(2)$

C_2H_4



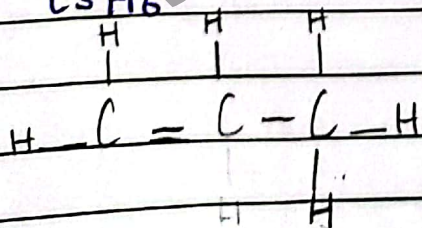
• Propene

C_nH_{2n}

$n = 3$

$\text{C}_3\text{H}_2(3)$

C_3H_6



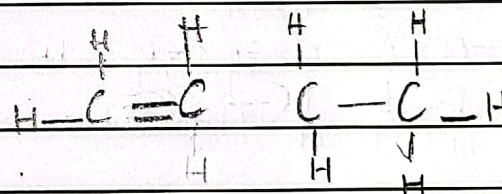
• Butene

C_nH_{2n}

$n = 4$

$\text{C}_4\text{H}_2(4)$

C_4H_8



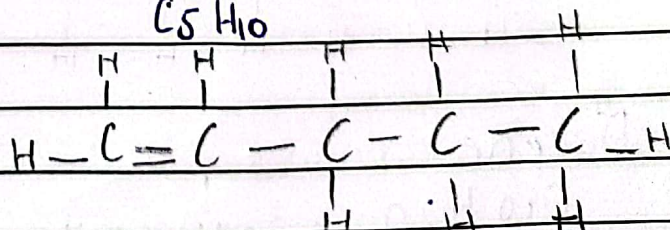
• Pentene

C_nH_{2n}

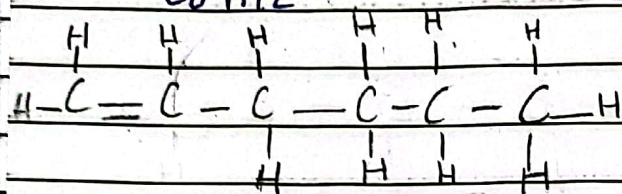
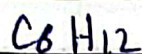
$n = 5$

$\text{C}_5\text{H}_2(5)$

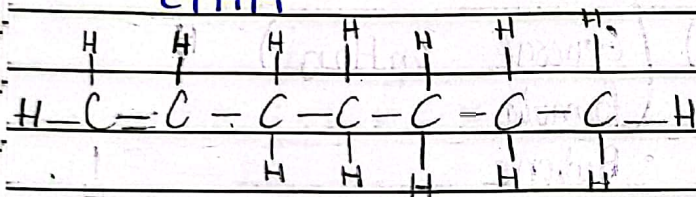
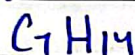
C_5H_{10}



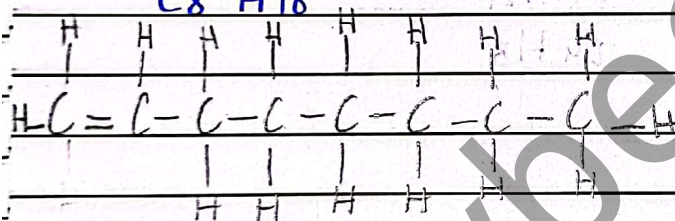
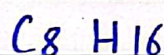
• Hexene



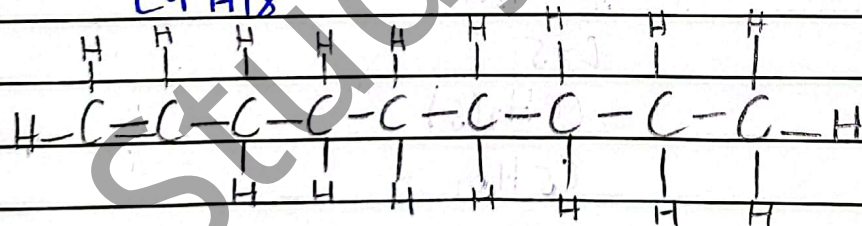
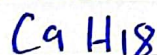
• Heptene



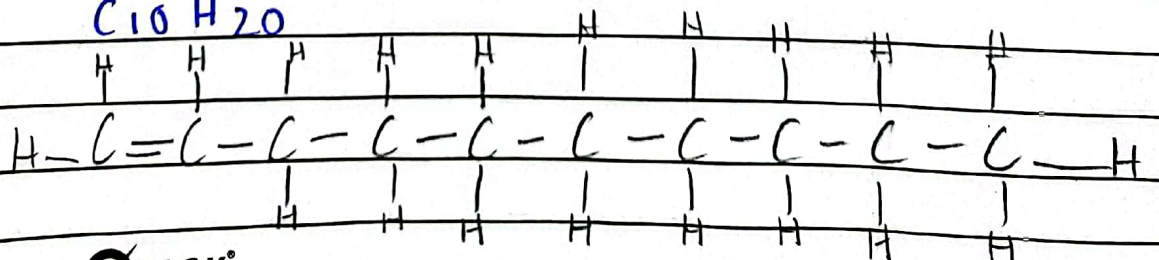
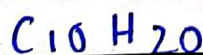
• Octene



• Nonene



• Decene



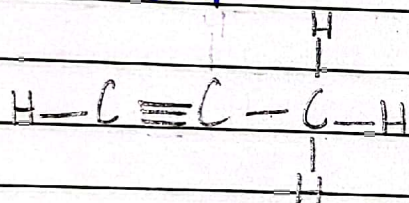
Alkynes

(Unsaturated) [General formula]
 $(C_n H_{2n-2})$

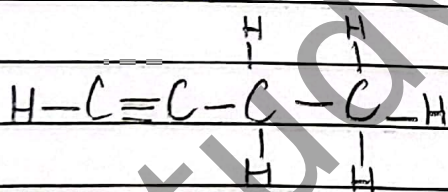
- Ethyne
 $C_2 H_{2(2)-2}$
 $C_2 H_{4-2}$
 $C_2 H_2$



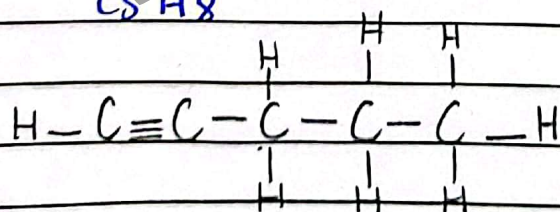
- Propyne
 $C_3 H_4$



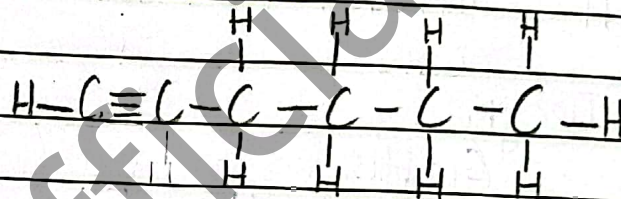
- Butyne
 $C_4 H_6$



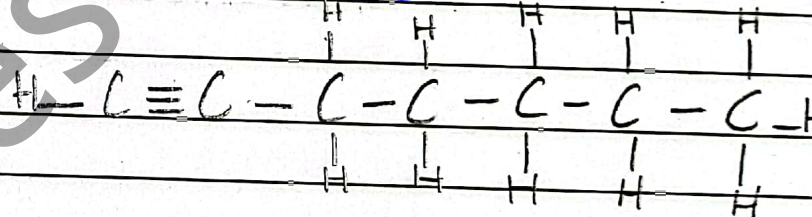
- Pentyne
 $C_5 H_8$



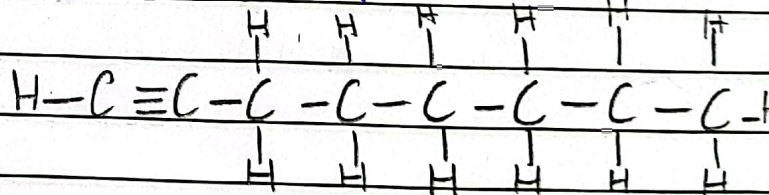
- Hexyne
 $C_6 H_{10}$



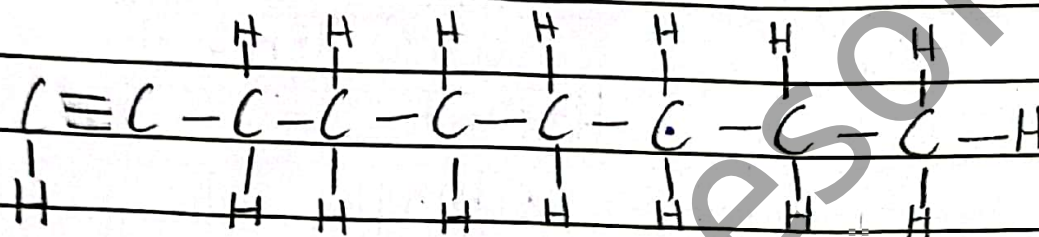
- Heptyne
 $C_7 H_{12}$



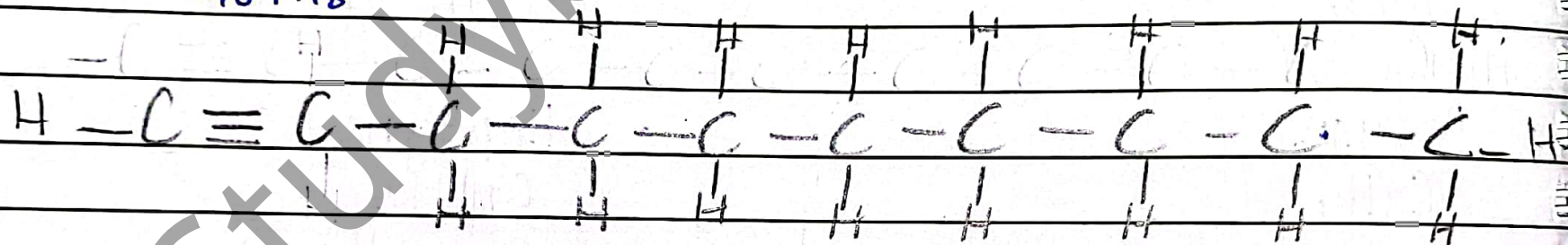
- Octyne
 $C_8 H_{14}$



• Nonyne
 $C_9 H_{16}$



• Decyne
 $C_{10} H_{18}$



Types of formulas :-

Molecular Formula :-

Shows the actual no. of each atom in a molecule.

e.g. :-



General Formula

Represents any member of a homologous series using n. (number of carbon atoms).

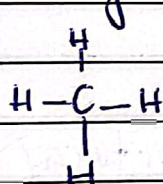
e.g. :-



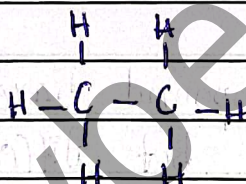
Structural formula

Describes the arrangement of atoms in a molecule.

e.g. :-



Methane



Ethane

Condensed structural form

A simplified version of structural formula where we don't draw all the lines and bonds - instead we write each carbon with its hydrogen as a group in sequence.

e.g. :-



Structural

vs

Condensed

Importance.

Bonds shown

Every bond drawn

No bonds drawn

In ethanol - O is attached to CH

How it looks

Full diagram with lines (bonds).

Written as groups

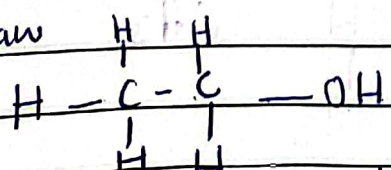
Dimethyl ether - O is bonded to 2 C-atoms.

Time to draw

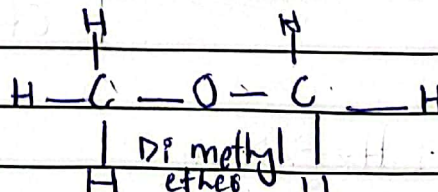
Takes longer

Quick n simple

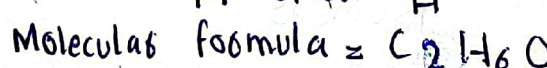
Same Molecular, totally diff structure, diff properties.



ethanol



Di methyl ether



Chemical Diversity

• Catenation :-

The self-linking ability of carbon to bond with other carbon atoms to form long chains, branched chains and rings.

- Carbon has 4 valence e⁻ → can form 4 bonds.
- No other element can compete carbon in this.
- Carbon chains can contain thousands of carbon atoms.
- Silicon can also catenate but forms only short chains.

• Isomerism :-

When two or more compound have the same molecular formula but different structural arrangements → Structural isomers.
 this phenomenon is called isomerism.

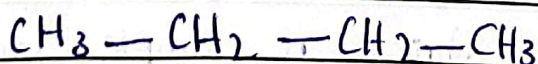
• Structural Isomerism :-

Straight chain - n

Carbons are connected in 1 continuous straight line.

e.g.:-

n-Butane (C₄H₁₀):-



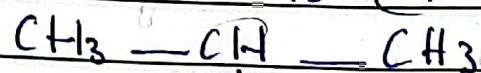
CH₄, C₂H₆, C₃H₈ → Non isomers

Branched chain - iso

Carbons have a branch / side chain coming off the main chain.

e.g.

iso-butane (C₄H₁₀)



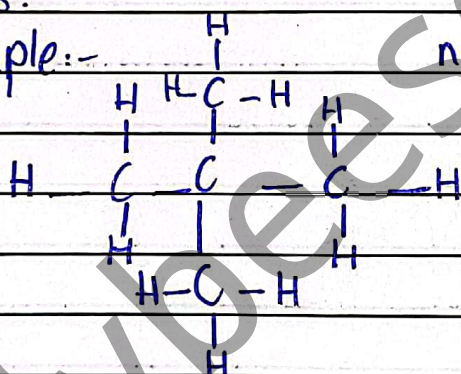
⇒ Chain isomerism begins with butane (C_4H_{10}) because compounds with fewer than four carbon atoms cannot be arranged in more than 1 way.

• Neo-Chain:-

→ highly branched chain (structure) with a carbon attached to four other carbons (a quaternary carbon).

→ Middle carbon is connected to four carbon groups.

Example:- neo-pentane (C_5H_{12})



• Remove 1 H from an Alkane → Alkyl group.

Alkane	Alkyl.
Methane CH_4	CH_3- Methyl
Ethane C_2H_6	C_2H_5- Ethyl
Propane C_3H_8	C_3H_7- Propyl
Butane C_4H_{10}	C_4H_9- Butyl

• Remove H_2 from an Alkane → Alkene

• Remove another H_2 from an Alkene → Alkyne.

Classification

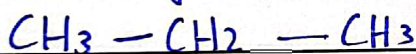
Organic compounds are broadly classified into 2 main groups based on their carbon skeleton:

→ Open chain or Acyclic compounds:-

These compounds contain an open chain of carbon atoms linked in a linear or branched pattern. They do not form rings.

For example :-

Straight chain



propane

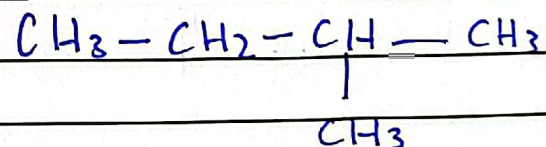


Butane

Branched chain



iso-butane



iso-pentane.

→ Closed chain or Cyclic Compounds :-

These are organic compounds that contain 1 or more rings of atoms.

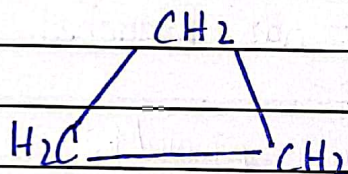
Homocyclic compounds

(Carbocyclic)

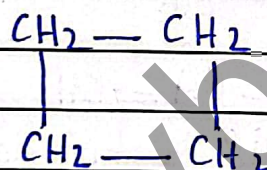
The ^{main} ring consist of only carbon atoms.

e.g. :-

Cyclopropane



Cyclobutane



Heterocyclic compounds

The main ring contain carbon atoms along with an atom of another element.

(like nitrogen, O, S, Sulphur)